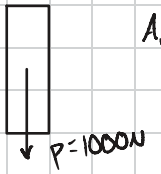
Leadscrew Minimum Diameter Using Axial Stress

Known: $P_{total} = 2kN = 2000N$
$n_{screws} = 2$
$P = \frac{2000}{2} = 1000N$
$SF = 4$
Leadscrew material: 1060 steel
$\sigma_y = 380\ MPa$

Unknown: Minimum root diameter (d_r)
Allowable Stress (σ_{allow})



$A_r = \frac{\pi d_r^2}{4}$

$\sigma_{allow} = \frac{\sigma_y}{SF} = \frac{380}{4} = \boxed{95\ MPa}$

$A_r = \frac{P}{\sigma_{allow}} = \frac{1000}{95\times10^6} = \boxed{1.053\times10^{-5}\ m^2}$

$d_r = \sqrt{\frac{4A_r}{\pi}} = \sqrt{\frac{4(1.053\times10^{-5})}{\pi}} = 0.00366m = 3.66mm \approx \boxed{0.144\ in}$ ½ ACME Leadscrew ✓

Rotational Speed

Known: $v_{min} = 1\ mm/min$
$v_{max} = 50mm/min$
$L = 2mm/rev\ (0.002m/rev)$

Unknowns: N_{min}
N_{max}

$V = L \cdot N \longrightarrow N = \frac{V}{L}$

$N_{min} = \frac{1\ mm/min}{2mm/rev} = \boxed{0.5\ rpm}$

$N_{max} = \frac{50mm/min}{2mm/rev} = \boxed{25\ rpm}$

Maximum Input Torque Required

Known: $P = 1000N$
$d = 0.5\ in = 0.0127m$
$L = 0.002m$
$d_m = d - \frac{L}{2} = 0.01143m$
Bronze friction coefficient: $f = 0.15$

$T = \frac{Pd_m}{2}\left(\frac{L+\pi f d_m}{\pi d_m - fL}\right) = \frac{(1000)(0.01143)}{2}\left(\frac{0.002+\pi(0.15)(0.01143)}{\pi(0.01143)-(0.15)(0.002)}\right)$

$\boxed{T = 1.21\ N\cdot m}$